METRIC
A-A-60057
27 November 2023
SUPERSEDING
A-A-52520B
28 May 2008

COMMERCIAL ITEM DESCRIPTION (CID)

HARDWOOD; FLOORBOARDS: FOR MILITARY VEHICLES, TRAILERS, & PLATFORMS (METRIC)

The General Services Administration (GSA) has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1 SCOPE. This CID covers engineered hardwood floorboards and floorboard kits fabricated for exterior cargo bed applications on vehicles, trailers, and platforms. Objectives of this CID are to provide an engineered Northern Red Oak domestic replacement for internationally recognized critically endangered tropical rainforest Apitong or Keruing (*Dipterocarpaceae dipterocarpus* spp.*)* floorboards and superseded CID A-A-52520. Foreign sourcing of now DoD-banned Apitong, critically endangered since 1998, was unsustainable, counter to numerous homologated U.S. Government regulatory/ethical commitments for responsible environmental stewardship in procurement, and non-compliant with the Trade Agreements (1979, 19 U.S.C. §§ 2501-2581) & Buy American (1933, 41 U.S.C. §§ 8301-8305) Acts. Northern Red Oak is classified as 'least concern for extinction' and is the main commercial species group of Oak lumber produced by the domestic hardwood industry, with abundant new growth greatly exceeding harvest throughout its native range of the eastern half of the continental U.S. Red Oak is strong, abrasion resistant, readily absorbs preservative treatment, is cost-effective, has much lower embodied transit CO_2 emissions than Malay Apitong, and has a prolific renewable-sustainable supply from a known homeland forest products supply chain/workforce. Apitong floorboards under superseded CID A-A-52520 were untreated and lacked biodeterioration resistance which often led to premature rot failure and elevated replacement life-cycle costs as a result. The Northern Red Oak floorboards under this CID will be preservative durability enhanced to better maintain mechanical properties over time for environmental exposure testing proven years longer life than Apitong.

2 SALIENT CHARACTERISTICS.

2.1 Materials. Unless otherwise specified herein, the materials used shall be in accordance with the manufacturer's material specification for hardwood. The use of recovered materials made in compliance with regulatory requirements is acceptable provided all requirements of this CID are met.

2.1.1 Hardwood. Floorboard base material shall be heartwood of the Northern Red Oak spp. grouping as defined by the Northeastern Lumber Manufacturers Association (NELMA) *Standard Grading Rules for Northeastern Lumber* and in accordance with MM-L-736, *Federal Specification: Lumber; Hardwood* Type II, Hardwood Construction Lumber. Surfaced 2 Sides (S2S) lumber shall be used. The use of trees grown in lowlands, wetlands, or swamps is unacceptable. Any lumber shall be excluded which contains accumulated defects to the extent that strength properties are decreased by an amount that threatens the

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FSC 2510

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end floorboard product application serviceability. Floorboard lumber shall be domestically grown, sourced, milled, finger jointed, & treated.

2.2 Phytosanitation & moisture content. The base constituent lumber segments used in floorboard construction shall meet the phytosanitary Kiln-Drying and Heat-Treatment (KD-HT) certification requirements specified in the U.S. Department of Agriculture (USDA) *Export Program Manual*. The moisture content of the finished manufactured and preservative treated floorboard product shall not exceed 20 percent ± 3 percent. Moisture content measurement shall be according to ASTM D7438 *Standard Practice for Field Calibration and Application of Hand-Held Moisture Meters*. Moisture content shall be verified using ASTM D4442 *Standard Test Method for Direct Moisture Content Measurement of Wood and Wood-Based Materials*. Wood will swell and shrink when its Moisture Content (MC) increases and decreases respectively when the MC is below 30%. Application board width tolerances will account for moisture content variability and will change up to 0.00287 per length unit multiplier per 1 percent MC change in the MC range of 0 percent to 30 percent.

2.3 Finger jointing. Floorboards shall be continuous for the required length without butt-joints or breaks. For the required continuous length floorboards, only solid lumber and uniform interlocking finger-jointed lumber in length is acceptable. Floorboard lumber joined in width is unacceptable. If longitudinal jointing is utilized, it shall be the structural horizontal configuration finger type. Due to having increased glue line length exposure with fingers subject to top surface weathering, vertical finger jointing of floorboard segments is unacceptable. Finger jointed lumber segments shall be a minimum of 1.22 m (48 inch) in length at the ends. Interior middle finger jointed lumber segments shall be a minimum of 2.28 m (90 inch) in length. During floorboard manufacture prior to jointing, a best effort should be made selecting the better constituent segment lumber faces for clarity and minimal defects facing up. Finger jointing adhesives shall be U.S. produced phenol-resorcinol-formaldehyde based that meet the exterior adhesive requirements specified in ASTM D2559 Standard Specification for Adhesives for Bonded Structural Wood Products for Use Under Exterior Exposure Conditions using a White Oak (Quercus alba) substrate. Finger jointing shall be per ISO 10983 *Timber – Finger Joints – Minimum Production Requirements and Testing Methods*, including: bending strength verification to a mean no less than 75% of the base wood (using the same specimen dimensions and test set up); overall production (as related to equipment, facilities, materials, personnel, procedures, processes, & etc.); and production batch quality conformance for initial qualification and ongoing manufacturing compliance testing. The bending strength of the glued finger joint shall be tested using the off-line test method specified in ASTM D7469 *Standard Test Methods for End Joints in Structural Wood Products*. The resistance of the bond to hydrothermal delamination shall be tested and evaluated according to American Institute of Timber Construction (AITC) T110 *Cyclic Delamination Test*. Test sampling by shift/line per ISO 10983 protocols of one representative finger joint for bending strength and one for hydrothermal delamination, shall be conducted at minimum for maintaining daily production quality compliance. Defects in wood component segments shall not impair the quality of joints. No knot, split, or similar compromising defect shall be permitted at a machined/glued finger joint. Knots shall be at minimum one knot diameter away from finger joints when measured in the longitudinal direction from the knot edge to the nearest finger joint edge.

2.4 Preservative treatment and end grain sealing. Floorboards shall be treated with U.S. produced, Environmental Protection Agency (EPA) registered, non-restricted (landfill safe), general-use, water-borne copper naphthenate wood preservative standardized by American Wood Protection Association (AWPA). The copper naphthenate used in the treating solution, suspension, or emulsion shall comply with AWPA P34 *Standard for Copper Naphthenate, Waterborne (CuN-W)*. Treatment shall be accomplished after all cutting, machining, trimming & finger jointing has been accomplished. Complete floorboards shall be treated with a low-odor water-borne copper naphthenate base preservative blend to the retention level (measured by gauge or weight) of AWPA Use Category UC4A (1.76 kg/m^3 of copper mass per unit wood volume) *Ground Contact (4), Commodity Specification (A) Sawn Products* as defined by the AWPA U1 *Use Category System: User Specification for Treated Wood*. Both pressure and non-pressure treatments are acceptable. In conjunction to the above, respective pressure and non-pressure treatment Quality Assurance shall be per AWPA T1 *Use Category System: Processing and Treatment Standard* and AWPA M21 *Standard for Quality Control Inspection of Nonpressure Preservative Treated Millwork Products*, as applicable. The CuN-W based treating solution, suspension, or emulsion shall also contain a minimum of 2 ± 0.1 mass percent concentration (% m/m) Disodium Octaborate Tetrahydrate (DOT), and 1 ± 0.1 % m/m Didecyl

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Dimethyl Ammonium Chloride (DDAC Quaternary). The above specified preservative blend provides protection against biological (mold, algae, fungus, & insect) attack and is corrosion inhibiting. If manufacturer damage occurs during material handling, stocking, or delivery loading that compromises preservative protection, the affected area(s) shall be repaired by surface treating liberally with the specified water-borne based copper naphthenate preservative application prior to final destination shipment. If further modification is required after treatment or during floorboard installation, any new exposed surfaces from machining, drilling, cutting, or trimming shall be liberally surface treated with the specified water-borne copper naphthenate preservative application. Following treatment and initial drying, to minimize future end checking and drying splits, the cut end-grain at each board terminus shall be liberally seal saturated till absorptive refusal with appropriate non-toxic aqueous-based petroleum wax emulsion coating.

2.5 Hardwood Defects. Lumber used in the fabrication of floorboard components shall conform to the below requirements of 2.5.1 through 2.5.8.

2.5.1 Fissures and Pockets. End splits exceeding the half board width in length, shall not be permitted; end splits shall only be allowed in one end of board. Open shakes, surface checks, end checks, and pockets wider than 3 mm (0.12 inch), exceeding on third thickness in depth, and half board width in length, shall not be allowed.

2.5.2 Hit-or-miss surfacing. Hit-or-miss surfacing, with skips not over 1.6 mm (0.06 inch) deep between them, shall be permitted on the underside of the floorboards, but all floorboards shall be full thickness at the ends and over the cross bolsters.

2.5.3 Knots. No unsound knot or hole shall be permitted. Sound knots will be permitted, up to and including 25.4 mm (1 inch) in diameter. The sum of the diameters of all sound knots in any one-fourth length of a piece shall not exceed the width of the piece, nor shall any two knots of 25.4 mm (1 inch) in diameter be closer to each other than 20 centimeters (cm) (8 inch).

2.5.4 Pith. No pith is permitted.

2.5.5 Twist. Maximum twist shall not exceed 1 mm (0.04 inch) per 30.5 cm (12 inch) length or equivalent ratio.

2.5.6 Wane. No wane, including sapwood, is permitted.

2.5.7 Warping. Maximum warping shall not exceed 1.5 mm (0.06 inch) per 30.5 cm (12 inch) length or equivalent ratio.

2.5.8 Wood destroying organism defects. Open channels not more than 3 mm (0.13 inch) deep and 12.7 cm (5 inch) long will be permitted on the face of the board. On the underside of the board, channels shall be not greater than 6 mm (0.25 inch) wide and 20 cm (8 inch) long. No defects shall extend through the thickness of the piece. No unsound degradative fungal decay is allowed.

2.6 Environmental exposure performance. The specified wood species treated with copper naphthenate to the AWPA Use Category UC4A *Ground Contact (4), Commodity Specification (A) Sawn Products* retention level shall be tested horizontally above ground in a mainland highly diverse biotically active deep tropical rainforest environment location within 6 degrees circles of latitude north and south from the equator by an accredited independent test lab that meets the requirements of International Organization for Standardization International Electro-technical Commission (ISO/IEC) 17025 *General Requirements for the Competence of Testing and Calibration Laboratories*. The tests shall include AWPA E16 *Standard Field Test for Evaluation of Wood Preservatives to be Used Above Ground (UC3B); Horizontal Lap-Joint Test.* The durability performance shall have a minimum average rating of 8.0 for decay and 9.5 for insects using the rating scale specified in AWPA E16 after 24 months of continuous environmental exposure.

2.7 Design and construction. Design, construction, and installation of floorboards shall conform to the applicable drawings and data supplied by the procuring activity (see 6.2). Finished engineered floorboard products shall meet the requirement of Select Grade or higher First & Seconds (FAS) or FAS 1-Face (F1F)

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as specified in the National Hardwood Lumber Association (NHLA) *Grading Rules*. Complete vehicular and trailer floorboard systems shall be compliant with the U.S. *Federal Motor Vehicle Safety Standards (FMVSS)* as codified in Title 49 Transportation of the Code of Federal Regulations, Part 571, Subtitle B Other Regulations Related to Transportation, Chapter III Federal Motor Carrier Safety Administration, Department of Transportation, Subchapter B Federal Motor Carrier Safety Regulations, Part 393 Parts and Accessories Necessary for Safe Operation § 393.84 *Floors*; "The flooring in all motor vehicles shall be substantially constructed, free of unnecessary holes and openings, and shall be maintained so as to minimize the entrance of fumes, exhaust gases, or fire. Floors shall not be permeated with oil or other substances likely to cause injury to persons using the floor as a traction surface." Boards shall be surfaced and of the required application profile dimensions. Standard floorboard tolerances, unless otherwise specified, shall be the following: length 0 mm / 5 mm (0.20 inch); width ± 2 mm (0.08 inch); thickness 0 mm / 2 mm (0.08 inch); lap thickness 0 mm / -2 mm (-0.08 inch); lap width 0 mm / 2 mm (0.08 inch). For floorboard kits, finger joints in boards adjacent to each other shall not be in the same drawing defined span between frame structure lateral crossmember cavities. No joints shall be on the frame lateral crossbars that may be compromised by drilling or fasteners. Floorboard fastening hardware and installation thereof shall be countersunk from the top board surface and specified in detail per end-item application engineering drawings [including, but not limited to requirements for; type/dimensions (i.e., carriage bolt or self-tapping metal screw), material, corrosion resistance, quantity, spacing & locations, installation torque, drilled hole dimensions & CuN-W repair treatment, etc.].

2.8 Quality impairment. Wood shall conform to stated requirements and shall be free from or imperfections to the extent that strength and serviceability is not decreased or otherwise detrimental to the required service of the component. Defects in wood components shall not impair the quality of the fitting of fastener hardware; no knot, split, or similar defect shall be permitted at a bolt, screw, machined glued finger joint, or notch lap, where it may impair the strength and serviceability of the component.

2.9 Identification and marking. Identification and marking of the lumber shall be clearly visible/legible with black 1 cm (0.40 inch) height bold letters or larger, and shall include, as a minimum: country of origin with hardwood type [**U.S. RED OAK]**; this CID number [**A-A-60057]**; phytosanitary Kiln Dried & Heat Treatment verification marking [**KD-HT**]; AWPA commodity specification, Use Category (UC), preservative code, and nominal minimum retention [**AWPA U1(A) UC4A P34 CuN-W RETEN 1.76 kg/m^3**]; Government Engineering Support Activity (ESA) CAGE code (i.e. 19207 for Army TACOM) with Applicable Engineering Drawings (AED) part and kit floorboard numbers (X) [**CAGE-AED-X**]; manufacturer's cage code [**MFR-CAGE**], National or North Atlantic Treaty Organization (NATO) Stock Number (NSN) [**NSN XXXX-XX-XXX-XXXX**]**,** manufacturer's brand mark & contact information, date of manufacture [**DATE MFR Month/Date/Year**], and arrow indicating floorboard installation front [↑ **FRONT** & **TOP** or **BOTTOM**]. Identification marks shall be duplicate located on the top and bottom surfaces, approximately 22.9 cm (9 inch) from the front installed end of the final application. The floorboards shall be permanently marked after jointing prior to treatment using non-fading durable lumber marking ink that, to the greatest extent possible, is compatible with the specified treatment, wood, long-term exterior exposure, and end usage application. Following treatment initial drying, the marked surface areas of the finished floorboard product shall be protected with a clear aqueous acrylic exterior-grade top coat containing an EPA registered fungicidal preservative (such as 'IPBC' 3-Iodo-2 propynyl butyl carbamate) that is compatible with the lumber marking ink and fresh treatment of the specified CuN-W preservative blend.

3 REGULATORY REQUIREMENTS. The offeror / contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with § 23.403 of the Federal Acquisition Regulation (FAR).

4 PRODUCT CONFORMANCE.

4.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to drawings, instruction manuals, legal requirements, rules, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance. All quality conformance testing data and results must be made available by the manufacturer to the government for inspection and/or review upon request.

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4.2 Responsibility for inspection. The contractor is responsible for the performance of all quality control inspections (examinations, tests, and detailed record keeping).

4.3 Condition of government production approval. As a condition of government approval prior to the contractor/manufacturer commencing Full Rate Production (FRP), procurement contract orders shall include provision for reverse engineered, modified, or newly designed complete floorboard kit system(s) referencing this CID to undergo an initial trial prototype installation on a platform weldment of the actual end-item application for validation inspection of technical configuration and quality contract requirements conformance. The prototype fit-up exercise will include technical participation from the floorboard kit contractor/manufacturer, the Engineering Support Activity (ESA) responsible for the end-item use application, and the government procurement contract management activity. The trial installation shall validate and initiate pre-production corrective action(s), if needed, for ESA Technical Data Package (TDP) documentation, to include; engineering drawings, Special Packaging Instructions (SPIs), & the utilized installation instructions.

4.4 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own defined drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects or non-conformances shall not be submitted for Government acceptance. The Government reserves the right to perform or witness any of the inspections it deems necessary to ensure supplies and services conform to prescribed requirements herein.

5 PACKAGING. Packing details shall be specified in SPI to be included with the Technical Data Package (TDP) referenced in the contract or order (see 6.2). As needed, SPIs will detail and specify requirements for; kit unit pack wood bundle support materials (blocking, bracing, cribbing, dunnage, means of long floorboards support), kit unit bundle stack pattern & board orientation (to prevent logistics & transit handling long floorboards finger joint bending damage), UV resistant protective weather storage covering, board bundle band strapping (material, locations, banding groove support lumber), overall marking (for shipping & storage of kit unit pack bundles; fork-lift points, identification information, weight, & dimensions), and packing wood phytosanitation/marking (International Plant Protection Commission IPPC International Standards for Phytosanitary Measures ISPM 15). If requested by foreign Customs for international order shipments of trailers or floorboard kits provided under this CID, the floorboard manufacturer shall coordinate with the USDA Animal and Plant Health Inspection Service (APHIS) and the U.S. Transportation Command (USTRANSCOM) in providing substantiating information to support generating the correct phytosanitary certificate documentation to successfully facilitate movement.

6 NOTES.

6.1 Source of documents.

a. AITC Standards: https://www.plib.org/aitc/resources/publications/design-standards/
b. ASTM International Standards: https://www.astm.org
c. AWPA Standards: https://awpa.com
d. Defense Logistics Agency Regulation Environmental Considerations in DLA Actions DLAR 1000.22: https://www.dla.mil/Portals/104/Documents/J5StrategicPlansPolicy/PublicIssuances/r1000.22.pdf
e. DoD Instruction, Natural Resources Conservation Program DoDI 4715.03: https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/471503p.pdf
f. DoD Instruction, Procurement of Sustainable Goods and Services DoDI 4105.72: https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/410572p.pdf
g. DoD Military Specifications & Standards: https://quicksearch.dla.mil/
h. DoD Sustainability Plan 2022: https://www.sustainability.gov/pdfs/dod-2022-sustainability-plan.pdf
i. Federal Acquisition Regulation (FAR): https://www.acquisition.gov/browse/index/far
j. GSA 2022 Sustainability Plan: https://www.sustainability.gov/pdfs/gsa-2022-sustainability-plan.pdf
k. IEC Standards: https://www.iec.ch/
l. ISO Standards: https://www.iso.org/

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m. International Union for Conservation of Nature (IUCN) Red List of Threatened Species: https://www.iucnredlist.org/resources/categories-and-criteriaiucnredlist.org
n. IPPC ISPM 15 Regulation of Wood Packaging Material in International Trade: https://www.aphis.usda.gov/aphis/ourfocus/planthealth/import-information/wood-packaging-material
o. NELMA Standard Grading Rules for Northeastern Lumber: https://www.nelma.org
p. NHLA Standards: https://nhla.com
q. Trade Agreements (1979, 19 U.S.C. §§ 2501-2581) & Buy American (1933, 41 U.S.C. §§ 8301-8305) Acts: https://uscode.house.gov/
r. U.S. Air Force Manuals for Environmental Management, Compliance, & Conservation 32-7001,2,3: https://www.e-publishing.af.mil/Product-Index/
s. U.S. Army DEVCOM Ground Vehicle Systems Center (GVSC) Environmental Policy: https://www.usarmygvsc.com/wp-content/uploads/2023/04/GVSC-Environmental-Policy_July2022.pdf
t. U.S. Army Environmental Regulation AR 200-1: https://armypubs.army.mil/ProductMaps/PubForm/AR.aspx
u. U.S. Army Tropic Regions Test Center: https://www.atec.army.mil/ypg/trtc.html
v. USDA Export Program Manual (Phytosanitary Certification Kiln Drying & Heat Treatment): https://www.aphis.usda.gov/import_export/plants/manuals/domestic/downloads/xpm.pdf
w. USDA Forest Service Climate Adaptation Plan FS-1196: https://www.usda.gov/sites/default/files/documents/4_NRE_FS_ClimateAdaptationPlan_2022.pdf
x. USDA Forest Service Forest Products Laboratory (FPL) Wood Handbook FPL-GTR-282: https://www.fpl.fs.usda.gov/documnts/fplgtr/fplgtr282/fpl_gtr282.pdf
y. U.S. Marine Corps Environmental Compliance and Protection Manual, MC Order P5090.2A: https://www.marines.mil/Portals/1/MCO P5090.2A W CH 1-3.pdf
z. U.S. Navy Environmental Readiness Program Manual OPNAV M-5090.1: https://www.navsea.navy.mil/Portals/103/Documents/SUPSALV/Environmental/5090.1.pdf
aa. U.S. CFR Title 49 Transportation § 393.84 Floors: https://www.ecfr.gov/current/title-49/subtitle-B/chapter-III/subchapter-B/part-393/subpart-G/section-393.84
bb. White House Executive Order 14005 Ensuring the Future Is Made in All of America by All of America's Workers: https://www.whitehouse.gov/wp-content/uploads/2021/06/M-21-26.pdf https://public-inspection.federalregister.gov/2021-02038.pdf
cc. White House Executive Order 14008 on Tackling the Climate Crisis at Home & Abroad: https://www.federalregister.gov/documents/2021/02/01/2021-02177/tackling-the-climate-crisis-at-home-and-abroad
dd. White House Executive Order 14057: Catalyzing Clean Energy Industries and Jobs Through Federal Sustainability: https://www.fedcenter.gov/programs/eo14057/
ee. White House Plan to Conserve Global Forests: Critical Carbon Sinks: https://www.whitehouse.gov/wp-content/uploads/2021/11/Plan_to_Conserve_Global_Forests_final.pdf

6.2 Ordering data. The contract or order should specify the following:

a. Title, Number, and date of this CID.
b. SPI selecting applicable levels of detailed packaging requirements.
c. Applicable drawings and data.

6.3 Definitions.

a. Critically Endangered: Considered to be facing an extremely high risk of extinction in the wild.
b. Decay: Deterioration of wood as a result of fungal attack. The typical appearance of decayed wood includes splitting across the grain, soft and punky, stringy, or crumbly. The presence of wood decay can result in significant reductions in strength. Decay must not exist in wood to be used for floorboards, as it has no strength and, under some conditions, may continue to spread into surrounding sound good wood.
c. Defect: Irregularity or imperfection in lumber that reduces the volume of sound wood or lowers its durability, strength, or utility value.
d. Delamination: Can be defined as separation of bond-line with poor or no bonding between adjacent glued layers, often caused by extreme changes in moisture content.
e. Durability: General term for permanence or resistance to deterioration of wood.
f. End Checking: Checking is a fissure of fibers across the annual rings in wood board ends as a result of stresses set up in wood during seasoning.

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g. End Grain: The grain of wood exposed when cut across the growth annual rings, perpendicular to fiber direction.
h. End Joint: Joint formed by adhesive bonding of machined, mated surfaces at the ends of two pieces of an engineered structural wood product as defined by ASTM D7469.
i. Environmental Ethics: The conceptual foundations of environmental values as well as more concrete issues surrounding human actions, policies, and the sustainable use of natural resources to protect biodiversity and maintain ecological systems.
j. Environmental Stewardship: Refers to the responsible use and protection of the biodiverse natural environment through active participation in conservation efforts and sustainable practices.
k. Environmental Sustainability: The responsibility to conserve natural forest resources and protect global ecosystems to support health and wellbeing values overall, now and in the future.
l. Face: Wide surface of rectangular pieces of lumber that is often the surface that determines grade.
m. Finger Joint: A finger joint is a woodworking joint made by cutting a set of complementary, tapered, interlocking profiles in two end pieces of wood, which are then glue bonded. The cross-section of the joint resembles the interlocking of fingers between two hands, hence the name "finger joint".
n. Finger Joint, Characteristic Strength: The characteristic bending strength, defined as the lower 5% tolerance limit strength with 75% confidence, determined by an appropriate statistical method, as per by ISO 10983.
o. Finger Joint, Compliance Testing: Testing in bending to verify ongoing production complies to characteristic strength established during qualification testing, as defined by ISO 10983.
p. Finger Joint, Horizontal: The finger profile appears only on the narrow thickness face of structural wood product.
q. Finger Joint, Production Batch: Group of finger joints, of the same base wood, adhesive, and nominal cross-sectional profile, made during a continuous production run, as defined by ISO 10983.
r. Finger Joint, Qualification Testing: Testing in bending to establish the lower 5% tolerance bending limit strength, with 75% confidence when first establishing the finger joint process, as defined by ISO 10983.
s. Finger Joint, Sampling: Representative finger joint samples for testing drawn at random from each work shift and production line, as defined by ISO 10983.
t. Fissure: Separation of wood fibers in a piece of lumber that can be classified in terms of its type, size, and location, i.e., shake, check, & end split.
u. Forest Products Supply Chain: Homeland providers, from acorn to floorboard, facilitating production and associated logistics of the end-product, includes forest lands owners, loggers, truckers, lumber mills, suppliers (for adhesive, treatment, packaging materials, etc.), & floorboard manufacturer.
v. Grade: Quality designation of lumber by application of a specified sorting procedure.
w. Hardwoods: Generally, trees of the botanical class *Angiospermae*, subclass *Dicotyledonae* that have broad leaves.
x. Heartwood: The more decay resistant inner core of the woody trunk stem wholly composed of non-living cells and differentiated for the outer enveloping layer (sapwood) by its darker color.
y. Kiln: A chamber used for the drying and conditioning of lumber in which the temperature and relative humidity of the circulated air can be varied and controlled.
z. Knot: A portion of a branch or limb which has been surrounded by subsequent growth of the wood of the tree.
   Sound Knot: A knot that is solid across its face, at least as hard as the surrounding wood and shows no indication of decay.
aa. Mechanical Properties of Wood: Structural characteristics that describe how wood behaves when subjected to external forces, i.e., to support loading or resist deflection.
bb. Moisture Content: The amount of water in wood usually expressed as a percentage of the weight of the wood. Moisture content is sensitive to changes in the surrounding environment. As the surrounding relative humidity increases, the wood moisture content will also increase, and as the surrounding relative humidity decreases the wood moisture content will also decrease.
cc. Off-Line Test: Test carried out on a sample taken from a continuous manufacturing process or from inventory of the product.
dd. Phytosanitation: Of, relating to, or being measures for plant diseases control, namely in forest products.
ee. Pith: Small, soft-core tissue occurring at the structural center of a log.

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ff. Pocket: Opening or defect between the rings of annual growth which develops during the growth of the tree and typically contains resin or bark, dark bark pockets being more typical to hardwoods.
gg. Preservative: Wood absorbed treatment products to increase durability by controlling and limiting biological wood degradation due to fungal rot or decay, molds, or wood-destroying insects.
Preservative retention: The nominal amount of preservative retained in the treated wood, as specified in AWPA Standard U1.
hh. Sapwood: Living tree portion at the time of harvest of the log wood in-between the exterior bark and inner heartwood. In general, sapwood is lighter in color and more susceptible to decay than heartwood.
ii. Shake: Longitudinal separation of wood.
jj. Sound wood: Free of any form of degradative decay, incipient or advanced, and from insect holes.
kk. Split: Separation of the wood parallel to the fiber direction, due to the tearing apart of wood cells.
ll. Spp.: Species pluralis; clarified plural abbreviation for multiple species (the most basic unit of classification and a taxonomic rank of an organism); several species under a genus (taxonomic rank of biological classification, comes above species and below family).
mm. Strength: The ability of a lumber member to sustain stress without failure.
nn. Structural Boards: Stress graded for structural applications for assignment of allowable properties.
oo. Surfaced 2 Sides (S2S): Lumber that is machine surface planned on the two widest sides to attain a smooth surface and uniform thickness, as defined by NELMA and NHLA.
pp. Thickness: The lesser dimension perpendicular to the long axis of lumber.
qq. Twist: Distortion caused by the winding of board edges such that the four corners of any face are no longer in the same plane.
rr. Use Category: System defined by the AWPA to provide a simplified categorization method of end-use biodeterioration hazard levels and expected performance of treated wood products.
ss. Wane: Lack of wood on any face or edge of a piece of sawn timber, including or not including bark.
tt. Warp: Variation from a true or plane surface of lumber.
uu. Wax Emulsion: Stable mixture of wax in water. Wax and water are normally incapable of forming a homogenous mixture but can be brought together stably using surfactants and other novel preparations.
vv. Weathering: Mechanical or chemical disintegration and discoloration of the surface of wood that is caused by UV light exposure, wind abrasion, and weather-related shrinkage and swelling.
ww. Wood Destroying Organism: Decay producing fungi, beetles, termites, and carpenter ants, that cause loss of strength properties in wood.

(For additional definitions, refer to ASTM D9 *Standard Terminology Relating to Wood & Wood Based Products.*)

CIVIL AGENCY

COORDINATING ACTIVITY:
USDA-OPS

MILITARY INTERESTS:

Custodian:
Army – AT
Air Force – 71
Navy - MC

Preparing Activity:
Army – AT

Review Activities:
Army - MD, MI
DLA – CC

(Project 2510-2024-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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